

Foundation Mathematics 1017SCG
Week 11 Workshop Questions

1. Evaluate the following integrals. (Hint: Don't forget the '+ C').

(a) $\int 4x^3 \, dx$

(b) $\int x^5 \, dx$

(c) $\int x - 2 \, dx$

(d) $\int 3x - 8 \, dx$

(e) $\int 5x^2 - 6x + 7 \, dx$

(f) $\int 9x^2 - 3x + 4 \, dx$

(g) $\int 7x^3 - 4x^2 + 9x - 2 \, dx$

(h) $\int 5 \sin(x) + 6 \cos(x) \, dx$

(i) $\int -4 \cos(x) + x \, dx$

(j) $\int 2e^x \, dx$

(k) $\int 6e^x + \sin(x) \, dx$

(l) $\int 7e^x + \frac{4}{x} \, dx$

(m) $\int -\frac{6}{x} + 2 \, dx$

2. Evaluate the following definite integrals.

(a) $\int_1^4 x \, dx$

(b) $\int_{-1}^3 x^3 \, dx$

(c) $\int_3^5 3x - 1 \, dx$

(d) $\int_{-2}^3 2x^3 \, dx$

(e) $\int_0^6 5 \, dx$

(f) $\int_{-1}^1 x^2 + 3x - 1 \, dx$

(g) $\int_{-2}^3 2 + 3x^2 \, dx$

- (h) $\int_0^{\frac{\pi}{2}} 3 \cos(x) \, dx$
- (i) $\int_{-2}^3 4e^x + 2x \, dx$
- (j) $\int_3^5 \frac{6}{x} \, dx$
- (k) $\int_0^2 2e^x - x^2 \, dx$
- (l) $\int_{-\pi}^{\frac{3\pi}{2}} -\sin(x) \, dx$
3. (a) Draw a reasonably accurate graph of $y = x^2 - 1$.
In the later parts of this question, we are interested in $0 \leq x \leq 2$ so it is important that this part of the graph is accurate.
- (b) Explain why $\int_0^2 x^2 - 1 \, dx$ is not equal to the area bounded by the curve $y = x^2 - 1$ and the x -axis between $x = 0$ and $x = 2$.
- (c) Evaluate $\int_0^2 x^2 - 1 \, dx$
- (d) Find the area bounded by the curve $y = x^2 - 1$ and the x -axis between $x = 0$ and $x = 2$.
4. Calculate the area bounded by the following curves and the the x -axis.
Hint: You may find it helpful to first sketch the functions.
- (a) $y = \sin(x)$ between $x = \frac{\pi}{2}$ and $x = \pi$.
- (b) $y = 2x + 3$ between $x = -1$ and $x = 2$.
- (c) $y = -x^2 - x + 6$ between $x = -2$ and $x = 1$.
- (d) $y = -x^2 + 4x$ between $x = -1$ and $x = 3$.
- (e) $y = \cos(x)$ between $x = 0$ and $x = \frac{3\pi}{2}$.
- (f) $y = \frac{1}{x}$ between $x = 1$ and $x = 3$.
5. In the first two minutes of mild exercise, a person's heart rate is given by $\frac{dB}{dt} = 4t^3 + 12t^2 + 75$ where t is the number of minutes since starting exercise and $0 \leq t \leq 2$. Each beat of the heart pumps 72 millilitres of blood.
- (a) Calculate the number of heart beats during the first minute of exercise.
Hint: If $\frac{dB}{dt}$ is the person's heart rate (number of heart beats per minute), how can we calculate the number of heart beats during the first minute?
- (b) Find the total amount of blood pumped during the first minute of exercise.
- (c) Find the total amount of blood pumped during the second minute of exercise.
- (d) What was the total amount of blood pumped during the two minutes of exercise?